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| **20MA2T02: DIFFERENTIAL EQUATIONS AND NUMERICAL METHODS** | | | | |

**COURSE OBJECTIVES:**

1. To enlighten the learners in the concept of differential equations.
2. To furnish the learners with basic concepts and techniques at plus two level to lead them into advanced level by handling various real world applications.

**COURSE OUTCOMES:**

At the end of the course, the student will be able to

**CO1:** Solve the differential equations related to various engineering fields (K3)

**CO2:** Identify solution methods of partial differential equations that model physical processes (K3)

**CO3:** Evaluate the approximate roots of polynomial and transcendental equations by different algorithms(K3)

**CO4:** Solve integrate and ordinary differential equations by various numerical techniques.(K3)

**CO5:** Solve ordinary differential equations by using different numerical schemes ( K3 ).

**Unit I: Linear differential equations of higher order:**

Solutions of Non-homogeneous equations of higher order with constant coefficients – with non-homogeneous terms of the type $e^{ax}$, sin ax, cos ax, polynomials in $x^n$, $e^{ax}$ V(x) and $x^n$V(x) – Method of Variation of parameters.

**Learning Outcomes:**

At the end of this unit, the student will be able to

- identify the essential characteristics of linear differential equations with constant coefficients (K3)
- solve the linear differential equations with constant coefficients by appropriate method (K3)

**Unit –II: Partial Differential Equations of First Order**:

Formation of partial differential equations by elimination of arbitrary constants and arbitrary functions – Solutions of first order linear (Lagrange) equations and nonlinear (standard types) equations.

**Learning Outcomes:**

At the end of this unit, the student will be able to

- apply a range of techniques to find solutions of standard PDEs (K3)
- outline the basic properties of standard PDEs (K2)

**Unit III: Interpolation**

Finite differences, Differences of a polynomial, relation between operators, to find one or more missing terms, Newton's interpolation formulae, and interpolation with unequal intervals- Lagrange's formula.

**Learning Outcomes:**

After the completion of this unit student will be able to

- explain various discrete operators and find the relation among operators ( K2 )
- apply Newton's forward and backward formulas for equal and unequal intervals ( K3 )

**Unit IV: Numerical Solution of Equations and Numerical integration**

**Numerical Solution of Equations:** Solution of algebraic and transcendental equations - Bisection Method, Method of False Position, Newton-Raphson Method, useful deduction from Newton-Raphson Method.

**Numerical Integration –** Trapezoidal rule, Simpson's $\frac{1}{3}$ rule and Simpson's $\frac{3}{8}$ rule.

**Learning Outcomes:**

After the completion of this unit student will be able to

- find approximate roots of an equation by using different numerical methods ( K3 )
- find integral of a function by using different numerical methods ( K3 )

**Unit V: Numerical Methods to Solve Ordinary Differential Equations**

**Numerical Methods to Solve Ordinary Differential Equations -** Taylor's series, Euler's and modified Euler's methods, Runge-kutta method of fourth order for solving first order equations.

**Learning Outcomes:**

After the completion of this unit student will be able to

- solve ordinary differential equations by using different numerical schemes ( K3 )

**Textbooks:**

1. B. S. Grewal, Higher Engineering Mathematics, 43/e, Khanna publishers,2015.

**References:**

1. Erwin Kreyszig, Advanced Engineering Mathematics, 9/e, John Wiley & Sons, 2013.
2. N.P. Bali and Manish Goyal, A text book of Engineering Mathematics, Laxmi Publications, 2008.